

PRAVIN KUMAR R

GEOSPATIAL DATA SCIENTIST

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PROFESSIONAL SUMMARY

Geospatial Data Scientist transitioning from an engineering analytical foundation (B.E. Civil Engineering, First Class Honors), specialized in multi-sensor Earth observation, remote sensing workflows, and deep spatial intelligence. Proven expertise curating and training models across optical imagery (13-band Sentinel-2 Level-2A), SAR (Sentinel-1 InSAR subsidence), and elevation profiles (Copernicus 30m DEM), achieving 95.98% land-cover accuracy and developing physics-grounded hydraulic disaster routing. Proficient in Python, PyTorch, Rasterio, GeoPandas, OpenCV, and Scikit-Learn, with European Space Agency (EO College) certification in Machine Learning for Earth Observation.

SKILLS

Earth Observation & Remote Sensing: Multi-Sensor Data Curation (Optical, SAR, DEM), Sentinel-2 (13 Bands, Level-2A Surface Reflectance), Sentinel-1 InSAR Subsidence, Copernicus 30m DEM, Atmospheric Correction, Orthorectification, Red Edge & SWIR Spectra

Geospatial Tooling & Data: Rasterio, GeoPandas, GDAL Basics, OpenCV, Spatial Coordinate Systems (CRS/WGS84), Tiling & Patch Extraction, Pandas, NumPy, SciPy, SQL, SQLite

Machine Learning & Spatial Vision: PyTorch, Swin-TransUNet, Convolutional Neural Networks (CNNs), Semantic Segmentation, Multi-Task Elastic Net, Random Forest, Monte Carlo Dropout (Spatial Uncertainty), Feature Engineering

Evaluation & Engineering Stack: Grounding & Generalization Benchmarks, Stratified Cross-Validation, Basel III Climate VaR Framework, Streamlit, Plotly, Pytest (Unit Testing & Spatial Code Gates), Git, GitHub, Agile

PROJECTS

Alveris-Geo-AI: Multi-Sensor Earth Observation & Climate Risk Engine

Python, PyTorch, 13-Band Sentinel-2, Copernicus 30m DEM, Sentinel-1 InSAR, Streamlit, Pytest

- Architected a multi-sensor Earth observation pipeline training a 13-band Sentinel-2 PyTorch CNN on Level-2A surface reflectance tensors, achieving **95.98% classification accuracy (+15.02% over 3-band RGB)** by resolving Red Edge and SWIR spectra.
- Modeled multi-decadal Sentinel-1 InSAR ground subsidence trajectories and engineered an 8-way morphological coastal flood routing algorithm on Copernicus 30m DEM, replacing planar bathtub models with physical breach hydraulics.
- Formulated an automated Basel III & SEC Climate Form 497 Value-at-Risk (VaR) framework translating spatial hazards into collateral haircuts; enforced reliability via **4 geodetic spatial code gates** and 49 passed pytest unit tests (PyLint 10/10).

HuBMAP: High-Resolution Semantic Segmentation & Spatial Vision Architecture

Python, PyTorch, Swin-TransUNet, Shifted Window Self-Attention, Monte Carlo Dropout, OpenCV

- Engineered an Organ-Prompted Swin-TransUNet combining shifted-window self-attention with spatial skip connections, achieving **0.6782 Dice and 0.5131 IoU (+26.7% over classical U-Net)** — architecture transferable to satellite land-cover mapping.
- Implemented Macenko optical density normalization using Beer-Lambert law and SVD matrix decomposition, eliminating cross-sensor spectral variance across large-scale high-resolution imagery.
- Constructed Monte Carlo Dropout inference to quantify pixel-level epistemic spatial uncertainty, generating calibrated confidence heatmaps to prevent false-positive boundary detections in downstream segmentation tasks.

Other Verified Projects: Soil Grain Size Estimation (55.22 OOF Weighted EMD, Multi-Task Elastic Net), CryptoCast (2.06% MAPE, PatchTST), ERIA Multi-Agent RAG (LLaMA 3.3 70B) – available on GitHub.

PROFESSIONAL EXPERIENCE

Data Science Intern — Internship Studio (Analytics & EdTech)

Jan 2026 – Feb 2026 | Remote

- Designed data ingestion, cleaning, and normalization pipelines for **14,102 wearable telemetry records**, achieving **99.8% feature completeness** under two-week Agile development sprints.
- Trained and evaluated a high-precision Random Forest regression model achieving **$R^2 = 0.9979$ and MAE of 3.78 kcal** (89.2% error reduction over baseline) to model nonlinear physiological exertion.
- Segmented user activity into 4 exertion zones using PCA and K-Means clustering (**Silhouette score 0.4546**), deploying an interactive Streamlit dashboard that cut reporting turnaround time by **75%**.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna Univ.

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: ML for Earth Observation (EO College), Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, Machine Learning & Pandas (HCL GUVI)

Languages: English (Professional), Tamil (Native)